Accelerated Gradient Methods

Computer Seminar, Week 4 (80 minutes)

DSBA

Plan and timing

Activity	Minutes
Lecture recap and setup	7
Quadratic trajectories and momentum	15
Stability map	10
Coordinate descent	10
Conjugate gradients	14
SGD and feature scaling	20
Discussion	4

Recap

For condition number $\kappa = L/\mu$:

GD: $O(\kappa \log(1/\varepsilon))$,

$$v_{k+1} = \beta v_k - \alpha \nabla f(x_k),$$
$$x^+ = x - L_i^{-1} \nabla_i f(x) e_i,$$

heavy-ball on quadratics: $O\left(\sqrt{\kappa} \log(1/\varepsilon)\right)$,

$$x_{k+1} = x_k + v_{k+1},$$
$$g_B = |B|^{-1} \sum_{i \in B} \nabla \ell_i(x).$$

CG solves positive definite quadratic problems through A-conjugate directions.

Experiment rules

Implement every optimizer using NumPy. Each method returns iterates, objective values, gradient or residual norms, and its work count. Compare methods using both iterations and primitive work such as gradient evaluations, coordinate derivatives, sample gradients, or matrix-vector products.

Task 1: elongated quadratic

Use

$$f(x) = \tfrac{1}{2} x^\top Q x, qquad Q = R^\top \operatorname{diag}(1, 100) R$$

with a nontrivial rotation R. Implement GD and Polyak momentum. Overlay their paths on identical contours and plot objective error against work.

Task 2: learning rate and momentum

For a grid of (α, β) values, run 150 heavy-ball iterations from one fixed point. Mark a pair stable when the final objective is finite and at least 10^3 times smaller than the initial objective. Plot a heatmap with three states: stable and fast, stable and slow, divergent.

Analytic stability check

For each eigenvalue λ, compute the roots of

$$r^2 - (1 + \beta - \alpha\lambda)r + \beta = 0.$$

The pair is linearly stable when the maximum root modulus over all eigenvalues is below one. Overlay this analytic boundary on the empirical map. Explain disagreements near the boundary using finite iterations and numerical tolerance.

Task 3: coordinate descent

Implement three selectors for a quadratic:

- cyclic coordinates;
- uniform random coordinates with a fixed seed;
- greedy choice maximizing $|\nabla_i f(x)|/\sqrt{L_i}$.

Use the exact coordinate step $-\nabla_i f(x)/Q_{ii}$. Compare objective error per coordinate update and per equivalent full pass.

Task 4: conjugate gradients

Implement CG for $Ax = b$, where A is symmetric positive definite. Stop on relative residual $\|r_k\| / \|b\|$. Record one matrix-vector product per iteration. Verify residual orthogonality and approximate A-conjugacy numerically.

Task 5: three solvers for one system

Solve the same system using

1. `numpy.linalg.solve`;
2. gradient descent on $\frac{1}{2}x^\top Ax - b^\top x$;
3. your CG implementation.

Compare solution error, relative residual, wall time, and matrix-vector products. State why the direct solver remains the accuracy reference for this small dense experiment.

Task 6: SGD without optimizer libraries

Generate a regression dataset and minimize mean squared error. Implement mini-batch SGD with shuffling, a final short batch, reproducible random state, and epoch-level logging. Compare constant, $1/\sqrt{k}$, and $1/k$ learning-rate schedules.

Task 7: feature standardization

Multiply input columns by strongly different scales. Train with the same SGD configuration before and after standardization. Compare the Hessian condition number, maximum stable learning rate found by a grid search, and loss against sample-gradient evaluations.

Important comparison

Convert standardized coefficients back to the original units before comparing parameter estimates.

Task 8: stochastic variability

Repeat SGD with at least five random seeds. Plot the median training curve with an interquartile band. Compare batch sizes 1, 16, and the full dataset while keeping the number of processed examples on the horizontal axis.

Expected deliverables

- implementations of GD, heavy-ball, coordinate descent, CG, and SGD;
- trajectory plot and heavy-ball stability map;
- solver comparison table with fair work measures;
- scaling experiment with condition numbers and recovered coefficients;
- brief explanations of momentum instability, CG efficiency, and SGD noise.

All cells should run from a fresh kernel.

Exit questions

Which curvature direction limits GD? Why does momentum need joint tuning of α and β? Why can a coordinate update be cheap? What property makes CG different from momentum? How does standardization change the optimization problem without changing model predictions?